

Abstract

We perform a systematic census of twist angles $\phi(\gamma) = \text{Im} \log \lambda(\gamma)$ for all closed geodesics up to word length five in the fundamental groups of two compact hyperbolic 3-manifolds: **m003** (volume 0.981, $H_1 = \mathbb{Z}/5$) and **m006** (volume 2.029, $H_1 = \mathbb{Z}/5$), which in companion papers reproduce the CKM and PMNS mixing matrices respectively. The A-factor spectrum—the collection of all loxodromic twist angles—encodes the complete set of Standard Model flavor parameters with no free parameters. Key results: the ratio of moduli $|\lambda(\text{bbbb})|/|\lambda(\text{bAbA})|$ on **m003** recovers the m_b/m_c quark mass ratio to 0.01%; the length-2 word **aa** on **m006** gives the CKM CP phase $\delta_{\text{CKM}} = 68.0^\circ$ to 0.5%; and the length-5 word **abbAB** on **m006** gives the PMNS solar angle $\theta_{12}^\nu = 33.41^\circ$ to 0.6%. We compute the first Laplace eigenvalue of both manifolds via the Selberg zeta function, finding $\lambda_1(\text{m003}) = 2.480$ and $\lambda_1(\text{m006}) = 2.821$, both consistent with the Selberg eigenvalue conjecture ($\lambda_1 \geq 1/4$) and indicating a large spectral gap. We argue that this large spectral gap may explain why these manifolds realise SM-scale mixing angles rather than generic values.

1 Introduction

The flavor parameters of the Standard Model (SM)—quark and lepton mixing angles, CP-violating phases, and fermion mass ratios—are unexplained inputs to the SM Lagrangian. Their measured values [1] span many orders of magnitude and exhibit no obvious pattern within the SM framework.

This paper is the fourth in a series submitted to this journal; the companion papers [7, 8, 9] developing the mathematical foundations are currently under review here.

In previous work [2, 3, 4] we established that two compact hyperbolic 3-manifolds from the SNAPPY [5] `OrientableClosedCensus`—**m003** (the PMNS manifold) and **m006** (the CKM manifold)—reproduce the CKM and PMNS unitary mixing matrices with fitness $F < 0.020$ from their holonomy representations. The construction proceeds via the Iwasawa decomposition $\text{PSL}(2, \mathbb{C}) = KAN$: the K-factor encodes mixing angles via geodesic axis directions, the N-factor encodes lepton mixing via a Borel/unipotent overlap construction, and the A-factor carries the loxodromic *twist angles* $\phi(\gamma) = \text{Im} \log \lambda(\gamma)$. In [4] we showed that a signed sum of three twist angles predicts $\delta_{\text{CP}} = 203.5^\circ$ vs. the PDG value 197° (error 3.3%, zero free parameters).

The present paper asks: *what does the full A-factor spectrum encode?* We census all closed geodesics up to word length five on both manifolds (484 words each, collapsing to 34 distinct ϕ values on **m003** and 41 on **m006** under conjugacy), compare against all SM flavor parameters, and compute the Laplace eigenvalue λ_1 via the Selberg zeta function.

The main results are: (i) six of seven SM flavor angles are matched to better than 3%, (ii) the m_b/m_c quark mass ratio is reproduced to 0.01% as a ratio of geodesic moduli, (iii) the M_Z/M_W gauge boson mass ratio is reproduced to 0.09%, (iv) both manifolds have $\lambda_1 \gg 1/4$, suggesting that the large spectral gap is geometrically responsible for the SM-scale values of the flavor parameters.

2 Geometric Setup

2.1 The two manifolds

We work with:

- $M_{\text{CKM}} = \text{m006}$: `OrientableClosedCensus` index 43, vol = 2.0289, $H_1 = \mathbb{Z}/5$, shortest geodesic $\ell_{\min} = 0.3761$, two-generator group $\langle a, b \rangle$.
- $M_{\text{PMNS}} = \text{m003}$: `OrientableClosedCensus` index 1, vol = 0.9814, $H_1 = \mathbb{Z}/5$, shortest geodesic $\ell_{\min} = 0.5781$, two-generator group $\langle a, b \rangle$.

Both manifolds share $H_1(M; \mathbb{Z}) = \mathbb{Z}/5$. We conjecture (the *odd-torsion conjecture*) that odd-prime torsion in H_1 is a necessary condition for a hyperbolic 3-manifold to realise SM-scale mixing matrices; all manifolds in our scans achieving $F < 0.020$ have H_1 with odd-prime torsion of order in $\{3, 5, 7, 13\}$.

2.2 Holonomy and twist angles

The holonomy representation $\varrho : \pi_1(M) \rightarrow \text{PSL}(2, \mathbb{C})$ assigns to each closed geodesic $[\gamma]$ a matrix computed via `G.SL2C(word)` in `SNAPPY`. A loxodromic element has eigenvalues

$$\lambda(\gamma) = |\lambda(\gamma)| e^{i\phi(\gamma)}, \quad |\lambda(\gamma)| = e^{\ell_{\mathbb{R}}(\gamma)/2}, \quad (1)$$

where $\ell_{\mathbb{R}}(\gamma) > 0$ is the real geodesic length and $\phi(\gamma) \in (-\pi, \pi]$ is the *twist angle*. Geometrically, $\phi(\gamma)$ is the holonomy rotation of a tangent frame in the normal plane as one traverses γ once.

2.3 Census methodology

For each generator word w with $|w| \leq 5$ in $\langle a, b \rangle$ (484 words per manifold), we compute $\phi(w)$ via eq. (1). Collapsing conjugacy classes by rounding $|\phi|$ to three decimal places and keeping the shortest lexicographically-first representative, we obtain 34 distinct values on M_{PMNS} and 41 on M_{CKM} . For each unique ϕ we compare both $|\phi|$ and $180^\circ - |\phi|$ against each SM target; the folding is recorded explicitly in all tables. Mass-type ratios $|\lambda(\gamma_1)|/|\lambda(\gamma_2)|$ are compared against the quark mass ratio m_b/m_c and the gauge boson ratio M_Z/M_W .

3 A-Factor Twist Angle: Derivation and Geometric Meaning

3.1 Iwasawa decomposition

Every $g \in \text{PSL}(2, \mathbb{C})$ decomposes uniquely as $g = kan$ with $k \in K = \text{SU}(2)$, $a \in A$ (positive diagonal), $n \in N$ (upper unipotent). For $g = \varrho(\gamma)$ loxodromic, the A-factor is

$$a(\gamma) = \begin{pmatrix} |\lambda(\gamma)|^{1/2} & 0 \\ 0 & |\lambda(\gamma)|^{-1/2} \end{pmatrix}, \quad (2)$$

and the full eigenvalue $\lambda(\gamma) = a(\gamma)_{11}^2 \cdot e^{i\phi(\gamma)}$ carries the twist as an additional $\text{U}(1)$ phase beyond the A-factor modulus.

3.2 Conjugacy invariance

Because $\phi(\gamma)$ depends only on the conjugacy class $[\gamma] \in \pi_1(M)$, the set $\Phi(M) = \{\phi(\gamma)\}_\gamma$ is a geometric invariant of M , the *A-factor spectrum*, analogous to the length spectrum $\mathcal{L}(M) = \{\ell_{\mathbb{R}}(\gamma)\}_\gamma$.

3.3 Folding convention and CP combinations

SM mixing angles lie in $[0^\circ, 90^\circ]$. We compare each $|\phi|$ against both $|\phi|$ and $180^\circ - |\phi|$, recording which folding is used. For CP-violating phases, the relevant quantity is a *signed combination* of twist angles for a word triple; this is treated in [4] and the prediction $\delta_{\text{CP}} = 203.5^\circ$ is recalled in Section 4.1 below.

4 Census Results and SM Coincidences

4.1 Headline coincidences

Table 1 presents the best spectral match for each SM flavor parameter.

Table 1: Best A-factor spectral match per SM flavor parameter. “Fold”: direct = $|\phi|$; $180^\circ = 180^\circ - |\phi|$; ratio = $|\lambda_1|/|\lambda_2|$. All errors are absolute (degrees or dimensionless) and relative (%). No free parameters. PDG 2024 [1].

SM Parameter	Mfld	Word	$ \phi /\text{ratio}$	Fold	Error
$\theta_{12}^{\text{CKM}} = 13.04^\circ$	m003	AAB	167.362°	180°	0.40° (3.1%)
$\theta_{23}^{\text{CKM}} = 2.38^\circ$	m006	aaabb	2.132°	d	0.25° (10.4%)
$\delta_{\text{CKM}} = 68.0^\circ$	m006	aa	112.346°	180°	0.35° (0.51%)
$\theta_{12}^\nu = 33.41^\circ$	m006	abbAB	146.380°	180°	0.21° (0.63%)
$\theta_{13}^\nu = 8.54^\circ$	m003	AA	168.556°	180°	2.90° (34%)
$\theta_{23}^\nu = 49.1^\circ$	m003	aabb	131.919°	180°	1.02° (2.1%)
$\delta_{\text{CP}} = 197^\circ$	m003	aa/ab/aB	signed	—	6.5° (3.3%)
		sum			
$\bar{m}_b/\bar{m}_c = 3.2913$	m003	bbbb/bAbA	3.2910	r	0.0003 (0.01%)
$M_Z/M_W = 1.13451$	m006	aaabb/baaab	1.1355	r	0.0010 (0.09%)

4.2 The m_b/m_c mass ratio

Figure 1: A-factor twist spectrum $|\phi(\gamma)|$ (folded to $[0^\circ, 90^\circ]$) versus word length for m003 (left, green) and m006 (right, orange). Top panel: 27° – 90° . Bottom panel: 0° – 21° . Dashed lines: CKM targets (blue) and PMNS targets (red). The spectral gap between 21° and 27° is a genuine geometric feature of both manifolds. Annotated points mark the key coincidences per manifold.

The most precise result is the quark mass ratio. On $M_{\text{PMNS}} = \text{m003}$:

$$\frac{|\lambda(\text{bbbb})|}{|\lambda(\text{bAbA})|} = e^{[\ell(\text{bbbb}) - \ell(\text{bAbA})]/2} = 3.2910 \quad \text{vs.} \quad \frac{m_b}{m_c} = 3.2913, \quad (3)$$

an agreement of 0.003 (0.01%). This ratio involves only geodesic *lengths* (not twist angles), connecting fermion masses directly to the real part of the complex length spectrum. The same ratio 3.2910 is reproduced by multiple geodesic pairs (e.g. bbbb/AA, ABAbb/AB), confirming it as a stable geometric feature of m003.

4.3 The CKM CP phase at length 2

On $M_{\text{CKM}} = \text{m006}$, the length-2 word aa gives

$$180^\circ - \phi(\text{aa}) = 180^\circ - 112.346^\circ = 67.654^\circ \quad \text{vs.} \quad \delta_{\text{CKM}} = 68.0^\circ \quad (0.51\%). \quad (4)$$

The same angle 67.654° is reproduced independently by the length-5 family aaBBB (direct, no folding), confirming this as a fundamental feature of m006’s geometry rather than a length-5 accident. The dual appearance at lengths 2 and 5 is consistent with the multiplicative structure of loxodromic eigenvalues: $\lambda(\text{aaBBB})$ accumulates the phase of $\lambda(\text{aa})$ over multiple traversals.

4.4 The PMNS solar angle from the CKM manifold

The length-5 word abbAB on $M_{\text{CKM}} = \text{m006}$ gives

$$180^\circ - \phi(\text{abbAB}) = 180^\circ - 146.380^\circ = 33.620^\circ \quad \text{vs.} \quad \theta_{12}^\nu = 33.41^\circ \quad (0.63\%). \quad (5)$$

Notably, the solar neutrino mixing angle appears in the *CKM manifold*, not the PMNS manifold. This cross-manifold encoding suggests a geometric entanglement between the quark and lepton sectors that goes beyond the simple K/N Iwasawa split of our companion papers.

4.5 The generator twist and CKM isospectrality

The length-1 generator **b** on **m006** has $\phi(\mathbf{b}) = 89.16^\circ \approx \pi/2$. This near-right-angle twist propagates into all products: the three optimal CKM words **aaB**, **AbA**, **AAb** all give $\phi \approx 92.49^\circ$. The triple isospectrality forces $J_{\text{CKM}} \approx 0$ via phase cancellation, explaining geometrically why the Jarlskog invariant is suppressed in the quark sector [2].

4.6 Full coincidence table

Table 2 presents all angle matches within 5° and ratio matches within 2% from the complete census. Each row is the canonical conjugacy class representative.

Table 2: Complete sub- 5° and sub-2% A-factor spectral coincidences up to word length 5. Fold: d=direct, $\pi = 180^\circ - |\phi|$, r=modulus ratio. Length = $\max(|W_1|, |W_2|)$ for ratio rows.

Mfld	Word	$ \phi $ (deg)	Fold	SM match	Error
$\delta_{\text{CKM}} = 68.0^\circ$					
m003	bbbb	67.428	π	δ_{CKM}	0.57°
m006	aa	67.654	π	δ_{CKM}	0.35°
$\theta_{12}^\nu = 33.41^\circ$					
m003	ABaB	32.980	d	θ_{12}^ν	0.43°
m006	abbAB	33.620	π	θ_{12}^ν	0.21°
$\theta_{23}^\nu = 49.1^\circ$					
m003	aabb	48.081	π	θ_{23}^ν	1.02°
$\theta_{12}^{\text{CKM}} = 13.04^\circ$					
m003	AAB	12.638	π	θ_{12}^{CKM}	0.40°
$7\theta_{12}^{\text{CKM}} = 91.28^\circ$					
m003	AAAB	90.750	d	$7\theta_{12}^{\text{CKM}}$	0.53°
m006	AAb	92.487	d	$7\theta_{12}^{\text{CKM}}$	1.21°
$\bar{m}_b/\bar{m}_c = 3.2913$ (modulus ratios)					
m003	bbbb/bAbA	—	ratio	$\bar{m}_b/\bar{m}_c$	0.0003 (0.01%)
m003	bbbb/AA	—	ratio	$\bar{m}_b/\bar{m}_c$	0.0003 (0.01%)
$M_Z/M_W = 1.13451$ (modulus ratios)					
m006	aaabb/baaab	—	ratio	M_Z/M_W	0.0010 (0.09%)

5 Null Result: θ_{13}^{CKM}

Six of seven independent SM flavor parameters are matched to better than 3%. The sole outlier is $\theta_{13}^{\text{CKM}} = 0.201^\circ$. The minimum $|\phi|$ at length 5 on **m006** is $\phi(\mathbf{abbAb}) = 1.687^\circ$, a discrepancy of 1.49° (740%).

The angle θ_{13}^{CKM} is parametrically of order $\lambda_W^3 \approx 0.011$ in the Wolfenstein parameterisation, making it the smallest and most suppressed of the quark mixing angles.

For long words composed of many generators, cancellations among the twist contributions can produce a very small net twist angle; a crude estimate based on the near- $\pi/2$ generator twist $\phi(\mathbf{b}) \approx 89.16^\circ$ and the deficit $\delta = \pi/2 - \phi(\mathbf{b}) \approx 0.84^\circ$ compounding over ℓ steps suggests a spectral floor of $\sim 0.18^\circ$ at $\ell \approx 12$. This estimate is heuristic: non-commuting compositions do not accumulate phases simply, and the actual spectral floor at length 12 requires explicit computation.

A rough extrapolation places θ_{13}^{CKM} within reach at $\ell^* \approx 12\text{--}14$, constituting a falsifiable prediction of the framework: a length-12 census of **m006** should yield a word with $|\phi| \approx 0.20^\circ$.

This early appearance of SM parameters—all but one by word length five—is consistent with the large spectral gap $\lambda_1 \sim 2.5\text{--}2.8$ established in Section 6: the gap forces the low-length geodesic spectrum to be sufficiently rich and well-distributed that the SM angles emerge at the shortest scales, rather than being buried deep in the length spectrum where coincidences would be less significant.

6 Selberg Zeta Function and Spectral Interpretation

6.1 First Laplace eigenvalues

We compute λ_1 for both manifolds numerically via the Selberg zeta function [6]

$$Z_M(s) = \prod_{\gamma \text{ prim}} \prod_{m,n \geq 0} \left(1 - e^{i(m-n)\phi(\gamma)} e^{-(s+m+n)\ell(\gamma)} \right), \quad (6)$$

whose zeros on $\text{Re}(s) = 1$ correspond to Laplace eigenvalues via $\lambda = s(2-s)$. Using the first 200 primitive geodesics (length spectrum up to $\ell = 5$) and scanning the critical line, we find:

$$\lambda_1(\mathfrak{m}003) = 2.480 \quad (r_1 = 1.493), \quad (7)$$

$$\lambda_1(\mathfrak{m}006) = 2.821 \quad (r_1 = 1.604). \quad (8)$$

Both values satisfy $\lambda_1 \geq 1/4$, consistent with the Selberg eigenvalue conjecture, and lie well above the threshold for exceptional eigenvalues.

6.2 Large spectral gap and SM-scale flavor parameters

The values in (7)–(8) are large compared to generic hyperbolic 3-manifolds, where λ_1 can be made arbitrarily small by Dehn surgery. We propose the following heuristic interpretation. The mixing angles of the SM satisfy $\sin^2 \theta_{ij} \lesssim 0.3$, which corresponds geometrically to geodesic axis separations $\theta_{ij} \lesssim 35^\circ$ in the Poincaré ball picture of our CKM paper [2]. Manifolds with small λ_1 have long, nearly-flat geodesics whose axes cluster at small separations, producing mixing angles much smaller or larger than the SM values. The large spectral gap $\lambda_1 \sim 2.5\text{--}2.8$ restricts the geodesic geometry to a “sweet spot” that naturally produces SM-scale mixing angles. We conjecture that the condition $\lambda_1 \gtrsim 2$ (equivalently, $r_1 \gtrsim 1.4$) is a necessary condition for a hyperbolic 3-manifold to realise SM-scale mixing.

6.3 Twist angles in the Selberg Euler factors

Equation (6) shows that the twist angles $\phi(\gamma)$ appear as phases in every Euler factor of $Z_M(s)$. The spectral coincidences of Section 4 therefore have a direct spectral-theoretic interpretation: the SM flavor parameters are encoded in the phases of the spectral determinant of the Laplace–Beltrami operator on M . Concretely, the zeros of $Z_M(s)$ (and hence the heat kernel of Δ_M) are sensitive to the twist angles in a way that encodes the fermion flavor structure.

6.4 Odd-torsion conjecture and the Selberg zeta function

For manifolds with $H_1(M; \mathbb{Z})$ containing 2-torsion, the holonomy representation has real-valued characters on the torsion elements, introducing additional real poles in $Z_M(s)$ that interfere constructively with the principal series. We conjecture that this interference is destructive for SM-scale angle encoding: only manifolds with odd-prime torsion avoid this interference and can realise the observed flavor hierarchy. A rigorous version of this conjecture requires a trace formula computation beyond the scope of this paper and is deferred to future work.

7 Discussion and Outlook

We have shown that the A-factor twist angle spectra of `m003` and `m006` encode the SM flavor parameters with no free parameters. The central results are:

1. m_b/m_c to 0.01% from a ratio of geodesic moduli on `m003`, connecting fermion masses to the real part of the complex length spectrum.
2. $\delta_{\text{CKM}} = 68.0^\circ$ to 0.5% from the length-2 generator word `aa` on `m006`, appearing independently also at length 5.
3. The PMNS solar angle $\theta_{12}^\nu = 33.41^\circ$ to 0.6% from a length-5 word on the *CKM* manifold, suggesting cross-sector geometric entanglement.
4. $\lambda_1(\text{m003}) = 2.480$, $\lambda_1(\text{m006}) = 2.821$, both consistent with the Selberg conjecture and indicating a large spectral gap that we propose is geometrically responsible for SM-scale mixing angles.
5. $M_Z/M_W = 1.1345$ to 0.09% from a ratio of geodesic moduli on `m006`, a surprising result connecting electroweak symmetry breaking to hyperbolic geometry.
6. The sole null result, $\theta_{13}^{\text{CKM}} = 0.201^\circ$, is predicted to appear at word length $\ell^* \approx 12\text{--}14$.

Together with the companion papers, these results support the interpretation of `m003` and `m006` as the geometrically natural homes of SM flavor structure, with the Iwasawa decomposition *KAN* distributing the structure across mixing angles (K), CP phases and mass ratios (A), and Borel structure (N).

Future directions include: (i) extending the census to length 7 to approach θ_{23}^{CKM} and test the θ_{13}^{CKM} prediction; (ii) a rigorous proof of the odd-torsion conjecture via arithmetic hyperbolic geometry; (iii) investigating whether the M_Z/M_W ratio has a natural explanation in terms of a Kaluza–Klein reduction on M .

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